

COMPANION SLIDES

Chapter 9

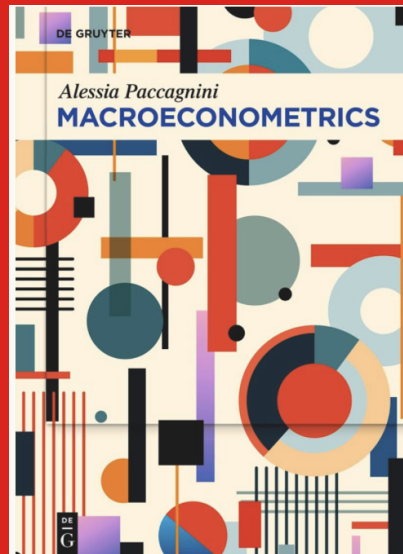
Machine Learning Tools

From regularization to nonlinear
forecasting

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Machine learning in macroeconomics

Can flexible methods improve forecasts when macroeconomic samples are short, dependent and structurally unstable?

- ▶ ML offers new tools for **prediction, variable selection and nonlinear relationships**.
- ▶ But flexibility creates a serious risk of overfitting.
- ▶ Interpretation matters when forecasts inform policy.

When does flexibility reveal signal rather than fit noise?

Foundations

- ▶ Four features of ML
- ▶ Bias–variance tradeoff
- ▶ Ridge, LASSO, Elastic Net
- ▶ Support Vector Regression
- ▶ Hyperparameter selection

Nonlinear and advanced methods

- ▶ Trees, bagging, random forests
- ▶ Gradient boosting
- ▶ Neural networks
- ▶ Advanced macro applications
- ▶ Interpretation and empirical evidence

- ▶ Explain why biased estimators can improve forecasts in high dimensions.
- ▶ Derive and distinguish Ridge, LASSO and Elastic Net.
- ▶ Use cross-validation for tuning parameters.
- ▶ Explain trees, bagging, random forests and boosting.
- ▶ Describe feedforward and recurrent neural networks and their regularization.
- ▶ Interpret variable importance, partial dependence and SHAP-style explanations.
- ▶ Evaluate ML forecasts with genuine out-of-sample designs and formal tests.
- ▶ Recognize when macro sample size and structural change limit ML gains.

1. Nonlinearity

How flexible is the mapping $g(Z_t)$?

2. Regularization

How strongly is complexity penalized?

3. Hyperparameter selection

How are tuning parameters chosen?

4. Loss function

What forecasting mistakes does estimation penalize?

$$\min_{g \in \mathcal{G}} \{ \widehat{L}(y_{t+h}, g(Z_t)) + \text{pen}(g; \tau) \}$$

$$\mathbb{E}[(y - \hat{f}(x))^2] = \sigma^2 + \text{Bias}[\hat{f}(x)]^2 + \text{Var}[\hat{f}(x)]$$

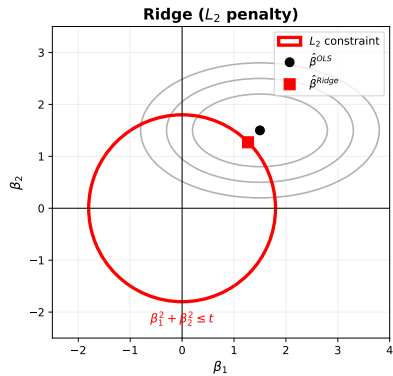
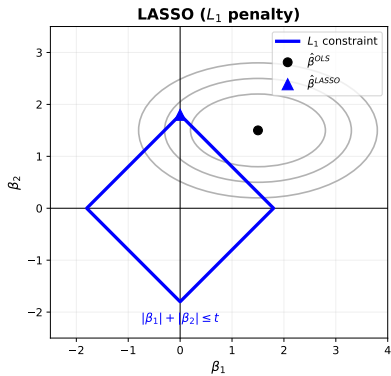
- ▶ Flexible models can reduce approximation bias but raise estimation variance.
- ▶ Macroeconomic samples are often small relative to the predictor set.
- ▶ Regularization deliberately accepts some bias to reduce variance.

Core idea

The best forecasting model need not be the least biased estimator.

$$\hat{\beta} = \arg \min_{\beta} \sum_{t=1}^T (y_t - X_t' \beta)^2 + \lambda \sum_{j=1}^p |\beta_j|^q$$

Method	Penalty	Main effect
LASSO	L_1	shrinkage + selection
Ridge	L_2	shrinkage
Elastic Net	$L_1 + L_2$	selection + grouping



$$\hat{\beta}^R = (X'X + \lambda I)^{-1}X'y$$

- ▶ Shrinks all coefficients toward zero but normally keeps every predictor.
- ▶ Stabilizes estimation under multicollinearity and even singular $X'X$.
- ▶ Useful when the macroeconomic DGP is **dense**: many small signals rather than a few dominant ones.
- ▶ Has a natural Gaussian-prior Bayesian interpretation.

$$\hat{\beta}^L = \arg \min_{\beta} \sum_t (y_t - X_t' \beta)^2 + \lambda \sum_j |\beta_j|$$

- ▶ Produces exact zeros and therefore variable selection.
- ▶ Attractive under genuine sparsity.
- ▶ Can be unstable when predictors are highly correlated.
- ▶ A sparse fitted model does not prove that the true macroeconomic DGP is sparse.

$$\text{pen}(\beta) = \lambda[\alpha\|\beta\|_1 + (1 - \alpha)\|\beta\|_2^2]$$

- ▶ $\alpha = 1$: LASSO; $\alpha = 0$: Ridge.
- ▶ Encourages correlated predictors to enter together: the **grouping effect**.
- ▶ Both λ and α are tuning parameters.

ε -insensitive loss

Errors inside an ε -tube receive zero loss; larger deviations are penalized.

- ▶ Kernels can introduce nonlinear relationships without explicitly constructing features.
- ▶ The approach combines regularization, a distinctive loss function and kernelization.
- ▶ Useful as another way to separate the contribution of **nonlinearity** from regularization.

- ▶ K -fold cross-validation balances fit and complexity.
- ▶ For time series, validation must respect chronology and avoid leakage.
- ▶ Pseudo-out-of-sample tuning is often more credible for forecasting.
- ▶ Hyperparameters should be chosen **inside** the forecasting exercise, not after seeing test performance.

Recursive partitioning

Split the predictor space into regions and assign a prediction to each terminal leaf.

- ▶ Naturally captures thresholds, nonlinearities and interactions.
- ▶ A single tree is interpretable but can have high variance.
- ▶ Tree depth, minimum leaf size and pruning control complexity.

Bagging

Fit trees on bootstrap samples and average predictions to reduce variance.

Random Forest

Also randomize the predictors considered at each split, reducing correlation across trees.

Ensemble diversity is useful when individual learners make different mistakes.

- ▶ Build weak learners **sequentially**, rather than independently.
- ▶ Each new tree focuses on errors left by the current ensemble.
- ▶ Learning rate and tree depth govern how aggressively the model adapts.
- ▶ Powerful, but tuning and validation are essential to prevent overfitting.

$$h^{(\ell)} = \phi(W^{(\ell)}h^{(\ell-1)} + b^{(\ell)})$$

- ▶ Hidden layers create flexible nonlinear representations.
- ▶ Depth and width increase approximation capacity.
- ▶ With macro data, the statistical challenge is often not representation but **sample size**.

Common tools

- ▶ weight decay
- ▶ dropout
- ▶ early stopping
- ▶ careful architecture choice

Time-series caution

- ▶ temporal dependence
- ▶ structural breaks
- ▶ data leakage
- ▶ unstable hyperparameters

- ▶ RNNs carry hidden states through time.
- ▶ LSTM and GRU architectures are designed to retain longer-range information.
- ▶ They can model nonlinear dynamics but require enough data to estimate reliably.
- ▶ In macroeconomics, benchmark comparisons remain essential.

- ▶ Large information sets can help identify policy shocks and summarize high-dimensional controls.
- ▶ ML can capture nonlinear responses or heterogeneous effects.
- ▶ Flexible nuisance estimation can be combined with econometric identification.
- ▶ Prediction quality alone does **not** establish causal identification.

- ▶ **Bayesian neural networks**: combine flexible functions with uncertainty quantification.
- ▶ **Tensor models**: exploit structured multidimensional/network information.
- ▶ **Quantile factor models**: connect dimension reduction with distributional and tail forecasting.

These methods extend the same themes: dimension, nonlinearity, regularization and uncertainty.

Evidence synthesized in the chapter

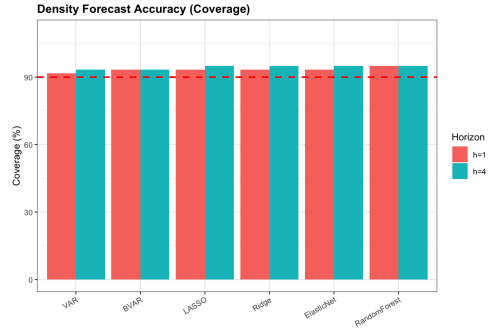
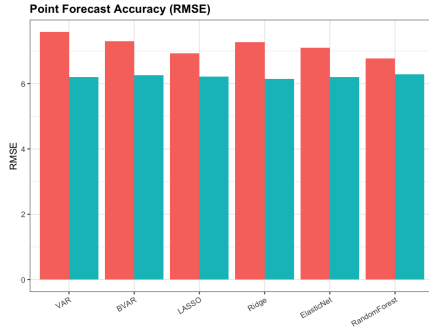
Forecasting gains are often linked more strongly to **nonlinearity** than to regularization alone.

- ▶ Gains can be state dependent.
- ▶ Stress and uncertainty may make nonlinear relationships more valuable.
- ▶ In tranquil periods, traditional models can remain highly competitive.
- ▶ No method dominates automatically across variables and horizons.

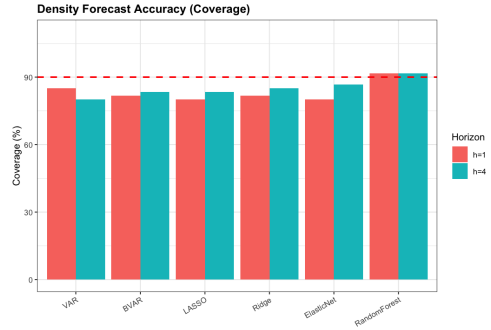
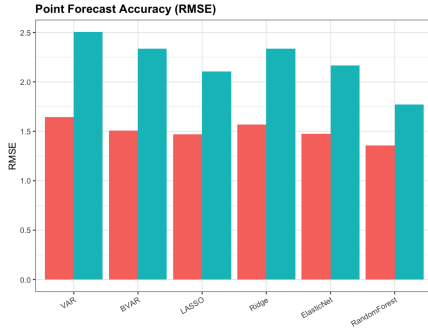
- ▶ Variable importance ranks predictors by contribution to fitted performance.
- ▶ Partial dependence summarizes how predictions vary with selected predictors.
- ▶ SHAP-style decompositions attribute individual predictions to features.
- ▶ Interpretation tools describe a fitted prediction mechanism; they should not automatically be read as causal effects.

- ▶ Data: real GDP, GDP deflator and Federal Funds rate.
- ▶ Sample: **1954Q4–2025Q3**, 284 quarterly observations.
- ▶ GDP growth and inflation: $400\Delta \log(X_t)$.
- ▶ Methods: VAR(4), BVAR(4), LASSO, Ridge, Elastic Net and Random Forest.
- ▶ Four lags of all three variables; direct forecasts.
- ▶ Expanding-window evaluation; final 60 observations reserved for OOS evaluation at $h = 1, 4$.

Forecasting Comparison: GDP Growth



Forecasting Comparison: Inflation

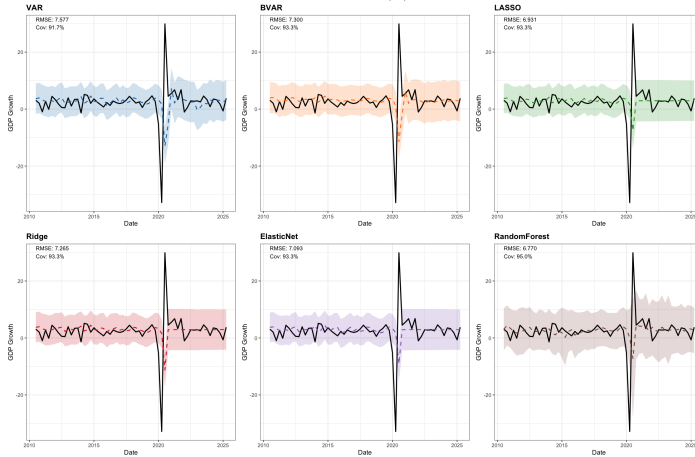


- ▶ GDP growth, $h = 1$: Random Forest has the lowest RMSE (6.77) versus 7.58 for VAR.
- ▶ GDP growth, $h = 4$: differences largely disappear; Ridge is marginally lowest at 6.14.
- ▶ Inflation, $h = 1$: Random Forest RMSE is 1.35 versus 1.65 for VAR.
- ▶ Inflation, $h = 4$: Random Forest RMSE is 1.77 versus 2.50 for VAR.

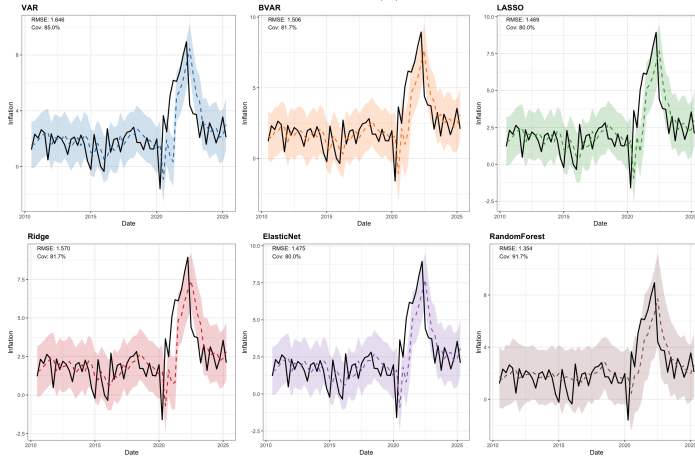
Interpretation in the chapter

The empirical gains are stronger for inflation and at horizons where nonlinear structure appears more useful.

GDP Growth Forecasts (h=1)



Inflation Forecasts (h=1)



- ▶ GDP growth coverage is broadly adequate, with wide bands influenced by the 2020 observations.
- ▶ Inflation is harder: most linear methods under-cover, especially at $h = 4$.
- ▶ Random Forest reaches 91.7% coverage for inflation at both horizons in the chapter exercise.
- ▶ The 2021–22 inflation surge illustrates how rapid regime change challenges all forecasting methods.

1. Define target, horizon and real-time information set.
2. Build credible econometric benchmarks.
3. Transform and standardize predictors without leakage.
4. Tune hyperparameters inside the training sample.
5. Generate genuine out-of-sample forecasts.
6. Compare point and density performance.
7. Use formal tests such as Diebold–Mariano where appropriate.
8. Interpret the fitted model and stress-test stability.

1. ML is not one method: nonlinearity, regularization, tuning and loss play distinct roles.
2. Bias can be useful when it buys a large reduction in variance.
3. LASSO selects; Ridge shrinks; ensembles exploit nonlinearities and interactions.
4. Macroeconomic sample size and structural instability constrain highly flexible models.
5. Forecast evaluation must remain genuinely out of sample and benchmark driven.

- ▶ Compare Ridge, LASSO and Elastic Net coefficient paths as λ changes.
- ▶ Replace ordinary K -fold CV with a time-series validation design.
- ▶ Compare a single regression tree, bagging and a random forest.
- ▶ Add gradient boosting or a small neural network to the empirical application.
- ▶ Compute variable importance or partial dependence for the inflation forecast.
- ▶ Repeat the exercise excluding or separately treating the COVID period.

Companion slides = starting point

These slides summarize core concepts and selected examples from Chapter 9. They are intentionally editable and are not a substitute for the full chapter.

- ▶ Add your preferred ML library and coding demonstrations.
- ▶ Add alternative macro/financial datasets and predictors.
- ▶ Add interpretability exercises and policy applications.
- ▶ Add new forecasting competitions, exercises and discussion questions.

Make them yours.

Nonlinearities

Regimes, time variation and state dependence

Next question

What if macroeconomic dynamics themselves change across regimes, states or over time?